COLREGS quick reference

Version 1.15 — the rules of the road, offline.

Commands

`colregs`	the menu
`colregs lights`	identify vessels by their lights
`colregs light <key> [brg]`	draw one, from any angle
`colregs shapes`	identify vessels by their day shapes
`colregs shape <key>`	draw one
`colregs sound`	identify sound signals
`colregs encounters`	who gives way, and what do you do
`colregs scenario`	a developing situation: plot it, decide, watch it run
`colregs rules`	the fifteen lessons
`colregs lesson L7`	one lesson
`colregs numbers`	which rule is which, both ways round
`colregs contacts`	bearing drift and contact management
`colregs track`	stand the tracking watch: drift, CPA, the call
`colregs ekelund`	range from a change of course
`colregs card`	the contacts card, on one screen
`colregs ref`	reference tables
`colregs style rn|usn|rel360|words|none`	how a relative bearing is spoken
`colregs day|night|plain`	colour mode
`colregs about`	why it exists, and how to send feedback

Bearings are relative to the other vessel. 0 means you are looking at her bow, 090 at her starboard side, 180 at her stern.

The lamp letters

W white **R** red **G** green **Y** yellow — beside every lamp, so the pattern reads with no colour at all. **fl** flashes, at 120 a minute or more, and the rate is part of the identification.

Lights worth knowing cold

Masthead, sidelights, sternlight	power-driven, under way
Two mastheads, after one higher	power-driven, 50 m or more
Sidelights and sternlight alone	sailing
Red over red	not under command
Red, white, red	restricted in her ability to manoeuvre
Red, red, red	constrained by her draught
Green over white	trawling
Red over white	fishing, other than trawling
White over red	pilot vessel on duty
All-round white, forward	at anchor, under 50 m

Sidelight cut-off

±2.5°, not exact. Annex I allows up to 3° outside the prescribed sector, so there is a small arc right ahead where **both** sidelights show. Do not call dead-ahead off one sidelight alone.

Bearing drift — the whole of it

> **Away from your bow:** she passes astern, or never crosses your track. **Toward your bow:** she crosses ahead, or draws ahead and clears.

No exceptions in 700,000 geometries. Relative motion is a straight line, so the bearing sweeps one way and cannot come back.

Good drift and still not safe — Rule 7(d)(ii):

1. a very large vessel — you plotted her bridge, her bow is nearer;
2. a tow — the tug clears, the barge on the wire does not;
3. close range — drift goes as the square of the range.

The report

Designation, bearing, drift, range, CPA, time. Bearing *before* drift. Say **“steady”** out loud — silence could mean you have not looked.

The arithmetic

Three-minute rule	yards travelled in 3 min = speed in knots × 100
Six-minute rule	miles in 6 min = speed in knots ÷ 10
CPA estimate	1° at 1 mile ≈ 35 yards
Ekelund	R(kyd) = 1.91 × ΔS⊥(kt) / Δω(°/min)
	S⊥ = own speed × sin(own course − bearing)

Spoken bearing styles

`rn`	“bearing 311, Red 20, drawing left”
`usn`	“bearing 311, Port 20, drawing left”
`rel360`	“bearing 311, 340 relative”
`words`	“bearing 311, fine on the port bow”
`none`	“bearing 311”

The true bearing is always given: it is what goes on the chart and what agrees with the radar.

What it will not do

1. It does not replace the Convention. Where they differ, the Convention is right.
2. It has no radar, no AIS and no position. Every exercise is generated.
3. It will not tell you that a situation is safe. It tells you what the rules require and what the geometry implies; Rule 2 is still yours.